AHOPPA: Agentic Humanoid Operation through Perception, Planning, and Action

Kingston Kuan[1] **Samuel J Chew**[2] **Brandon Ong**[1]
[1]Hebbian Robotics [2]Carnegie Mellon University

https://hebbian-robotics.github.io/AHOPPA/

Abstract: We study extended humanoid tasks with results from the last few weeks following frontier model releases. AHOPPA connects multimodal models to fixed SONIC control on a simulated Unitree G1 through visual feedback and numerical motion targets, without task skills. The model inspects four drawers, pushes open ones shut and returns home while physics continues during inference. Across 64 episodes with four models at two reasoning settings, GPT-6 Astra succeeded in 12/14 task attempts, taking 615 to 1492 s. Evaluation separates physical completion from declared success, exposing falls, timeouts and false declarations. Higher effort sometimes improved physical progress without improving success.

Keywords: Humanoid control, multimodal agents, online evaluation

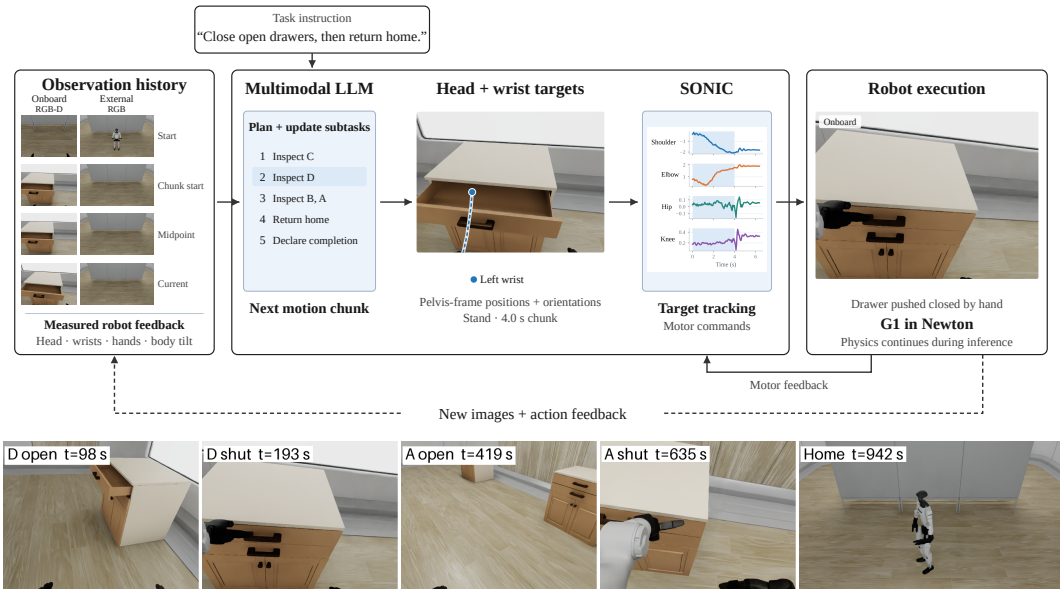


Figure 1: Agent loop and original observations from a successful Astra low episode with A and D open. Home observation: 942 s; declaration: 955 s.

1 Online humanoid operation

Room-scale humanoid operation couples navigation, inspection, contact and stopping while physics continues during inference. We test frontier models' ability to operate a humanoid through numerical references to the fixed SONIC controller [1]. Prior work spans skills [2, 3, 4], code [5, 6], visual feedback [7, 8], spatial representations [9, 10, 11] and learned action representations [12]. Our approach uses numerical commands without predefined task skills, generated policy code, value maps or keypoint constraints.

2 Interface, task and evaluation

Interface. Inputs are onboard RGB-D, external RGB, robot kinematics, depth queries, a map, the plan and feedback (Fig. 1). SONIC's gamepad commands set walking speed, direction, facing and posture; three-point tracking follows head/wrist positions and orientations. Four pose keyframes span a chosen duration; hand presets are optional. Drawer states, contacts and scores are withheld.

Task. Inspect four top drawers, push open ones shut by hand, leave closed ones undisturbed, return home and declare completion. Four passive Sektion drawers [13] occupy Simple Room [14, 15]: A/B behind lockers [16], C/D behind home; all fronts are initially hidden. Eight initializations: none (control), A, B, C, D, A+B, A+D or B+C open; all open drawers start at 0.20 m under the same instruction. Newton [17] in Isaac Lab [18] runs at 200 Hz.

Evaluation. Success requires all drawers within 2 cm of shut at $\leq$1 cm/s for 1 s, a supported standing hold within 25 cm of home and a completion declaration. Inspection, hand attribution and disturbance are scored separately. After pilots, we froze the setup and 1800 s budget. GPT-6 Astra, GPT-6.1 Sol, Claude Opus 5.5 and Gemini 4 Argon ran once per condition/effort, four concurrently.

3 Results

Model / effort	Success	Physical completion	False declarations	Falls	Mean API cost ($)	Episode time (s)	Decision latency (s)
Astra / low	6/7	6/7	0/7	1/7	9.30 / 10.57	817 / 926	9.2
Astra / high	6/7	6/7	0/7	1/7	7.65 / 8.78	998 / 1145	21.4
Sol / low	4/7	4/7	1/7	2/7	1.72 / 1.96	829 / 940	10.4
Sol / high	3/7	5/7	0/7	2/7	1.74 / 1.59	1235 / 1162	21.5
Opus / low	2/7	3/7	2/7	3/7	4.78 / 6.23	762 / 1028	13.2
Opus / high	3/7	3/7	1/7	2/7	7.22 / 8.36	1392 / 1563	36.1
Gemini / low	0/7	0/4	0/4	3/4	1.29* / –	737 / –	32.3
Gemini / high	0/7	0/5	0/5	1/5	1.59* / –	1332 / –	136.7

Table 1: Seven scheduled task attempts per row. Physical completion: goal reached at least once; false declarations: goal unmet when claimed. API errors remain in success denominators only. Gemini: three low-effort filter rejections, two high-effort timeouts. Cost/time: all/successful means through termination. Costs: usage at published rates [19, 20, 21]; GPU costs excluded; * excludes unresolved usage; –: no success. Latency: median of per-attempt medians for completed decisions.

Performance. Astra led (6/7 each); 11/12 successes viewed all four drawer fronts. Sol followed with 7/14 successes, Opus with 5/14, and Gemini with none (Table 1). All 43 closures were by hand; no initially closed drawer was disturbed.

High vs. low effort. Higher effort increased decision latency roughly 2–4$\times$; Astra's success was unchanged and Opus improved. Sol high closed more open drawers (9/10 vs. 4/10) and physically completed more tasks (5/7 vs. 4/7), but two went unreported before timeout or fall, yielding fewer successes (3/7 vs. 4/7). Another Sol high run fell before any model command.

Falls. Of eight all-closed controls, three succeeded, four fell and one had an API error. Overall, 16/19 falls lacked recorded drawer contact; other collisions remain possible. Eleven of 18 post-command falls ended in holds commanding zero translational speed with retained pose/facing targets. We hypothesize that prolonged target holds during inference contribute to instability.

Cost. Astra high matched low's success at 18% lower mean API cost; cached reads covered 50% of input versus 29% at low, despite more output tokens. Successful runs cost less for Sol high ($1.59 vs. $1.96), but including failures, Sol low spent less per success ($3.00 vs. $4.06).

Limitations. One layout and one trial per model/effort/condition limit generalization; 41/58 completed episodes exceeded 100 ms peak physics lag. Reliability and transfer need repeated trials and hardware tests.

References

[1] Z. Luo, Y. Yuan, T. Wang, C. Li, F. Castañeda, S. Chen, Z.-A. Cao, J. Li, D. Minor, Q. Ben, J. Park, D. Sami, Z. Wang, X. Da, R. Ding, C. Hogg, L. Song, E. Lim, E. Jeong, T. He, H. Xue, W. Xiao, S. Yuen, J. Kautz, Y. Chang, U. Iqbal, L. Fan, and Y. Zhu. SONIC: Supersizing motion tracking for natural humanoid whole-body control. *Science Robotics*, 11(117):eaed4592, 2026. doi:10.1126/scirobotics.aed4592. URL https://nvlabs.github.io/GEAR-SONIC/.

[2] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, C. Fu, K. Gopalakrishnan, K. Hausman, A. Herzog, D. Ho, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, E. Jang, R. J. Ruano, K. Jeffrey, S. Jesmonth, N. J. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, K.-H. Lee, S. Levine, Y. Lu, L. Luu, C. Parada, P. Pastor, J. Quiambao, K. Rao, J. Rettinghouse, D. Reyes, P. Sermanet, N. Sievers, C. Tan, A. Toshev, V. Vanhoucke, F. Xia, T. Xiao, P. Xu, S. Xu, M. Yan, and A. Zeng. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances, 2022. URL https://arxiv.org/abs/2204.01691.

[3] G. Huh, C. Gu, T. E. Truong, C. K. Liu, and G. Tevet. HomeBody: A humanoid that explores, remembers, and acts on its own, 2026. URL https://tml.stanford.edu/homebody/.

[4] J. Sun, Q. Zhang, Y. Duan, X. Jiang, C. Cheng, and R. Xu. Prompt, Plan, Perform: LLM-based Humanoid Control via Quantized Imitation Learning, 2024. URL https://arxiv.org/abs/2309.11359.

[5] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng. Code as Policies: Language Model Programs for Embodied Control, 2023. URL https://arxiv.org/abs/2209.07753.

[6] S. Choi, S. Ye, N. Bui, A. Shrivastava, K. Ryu, D. Tirumala, M. Wulfmeier, and N. Mehr. HuGo: LLMs as whole-body policy code designers for humanoid loco-manipulation, 2026. URL https://arxiv.org/abs/2609.30594.

[7] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar, P. Sermanet, N. Brown, T. Jackson, L. Luu, S. Levine, K. Hausman, and B. Ichter. Inner Monologue: Embodied Reasoning through Planning with Language Models, 2022. URL https://arxiv.org/abs/2207.05608.

[8] S. Nasiriany, F. Xia, W. Yu, T. Xiao, J. Liang, I. Dasgupta, A. Xie, D. Driess, A. Wahid, Z. Xu, Q. Vuong, T. Zhang, T.-W. E. Lee, K.-H. Lee, P. Xu, S. Kirmani, Y. Zhu, A. Zeng, K. Hausman, N. Heess, C. Finn, S. Levine, and B. Ichter. PIVOT: Iterative visual prompting elicits actionable knowledge for VLMs. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 37321–37341. PMLR, 2024. URL https://proceedings.mlr.press/v235/nasiriany24a.html.

[9] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models, 2023. URL https://arxiv.org/abs/2307.05973.

[10] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suenderhauf. SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 23–72. PMLR, 2023. URL https://proceedings.mlr.press/v229/rana23a.html.

[11] W. Huang, C. Wang, Y. Li, R. Zhang, and L. Fei-Fei. Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 4573–4602. PMLR, 2025. URL https://proceedings.mlr.press/v270/huang25g.html.

[12] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, Q. Vuong, V. Vanhoucke, H. Tran, R. Soricut, A. Singh, J. Singh, P. Sermanet, P. R. Sanketi, G. Salazar, M. S. Ryoo, K. Reymann, K. Rao, K. Pertsch, I. Mordatch, H. Michalewski, Y. Lu, S. Levine, L. Lee, T.-W. E. Lee, I. Leal, Y. Kuang, D. Kalashnikov, R. Julian, N. J. Joshi, A. Irpan, B. Ichter, J. Hsu, A. Herzog, K. Hausman, K. Gopalakrishnan, C. Fu, P. Florence, C. Finn, K. A. Dubey, D. Driess, T. Ding, K. M. Choromanski, X. Chen, Y. Chebotar, J. Carbajal, N. Brown, A. Brohan, M. G. Arenas, and K. Han. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR, 2023. URL https://proceedings.mlr.press/v229/zitkovich23a.html.

[13] Isaac Lab Project Developers. *Available Environments: Isaac-Open-Drawer-Franka-v0*, 2026. URL https://isaac-sim.github.io/IsaacLab/main/source/overview/environments.html.

[14] NVIDIA. *Environment Assets*. NVIDIA, 2026. URL https://docs.isaacsim.omniverse.nvidia.com/latest/assets/usd_assets_environments.html.

[15] NVIDIA. *NVIDIA Isaac Sim Additional Software and Materials License*. NVIDIA, 2025. URL https://www.nvidia.com/en-us/agreements/enterprise-software/isaac-sim-additional-software-and-materials-license/.

[16] NVIDIA. *SimReady Assets*. NVIDIA, 2026. URL https://huggingface.co/datasets/nvidia/simready-assets.

[17] The Newton Contributors. Newton: GPU-accelerated physics simulation for robotics and simulation research, 2025. URL https://github.com/newton-physics/newton.

[18] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrügg, N. Rudin, L. Wawrzyniak, M. Rakhsha, A. Denzler, E. Heiden, A. Borovicka, O. Ahmed, I. Akinola, A. Anwar, M. T. Carlson, J. Y. Feng, A. Garg, R. Gasoto, L. Gulich, Y. Guo, M. Gussert, A. Hansen, M. Kulkarni, C. Li, W. Liu, V. Makoviychuk, G. Malczyk, H. Mazhar, M. Moghani, A. Murali, M. Noseworthy, A. Poddubny, N. Ratliff, W. Rehberg, C. Schwarke, R. Singh, J. L. Smith, B. Tang, R. Thaker, M. Trepte, K. V. Wyk, F. Yu, A. Millane, V. Ramasamy, R. Steiner, S. Subramanian, C. Volk, C. Chen, N. Jawale, A. V. Kuruttukulam, M. A. Lin, A. Mandlekar, K. Patzwaldt, J. Welsh, H. Zhao, F. Anes, J.-F. Lafleche, N. Moënne-Loccoz, S. Park, R. Stepinski, D. V. Gelder, C. Amevor, J. Carius, J. Chang, A. H. Chen, P. de Heras Ciechomski, G. Daviet, M. Mohajerani, J. von Muralt, V. Reutskyy, M. Sauter, S. Schirm, E. L. Shi, P. Terdiman, K. Vilella, T. Widmer, G. Yeoman, T. Chen, S. Grizan, C. Li, L. Li, C. Smith, R. Wiltz, K. Alexis, Y. Chang, D. Chu, L. J. Fan, F. Farshidian, A. Handa, S. Huang, M. Hutter, Y. Narang, S. Pouya, S. Sheng, Y. Zhu, M. Macklin, A. Moravanszky, P. Reist, Y. Guo, D. Hoeller, and G. State. Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning, 2025. URL https://arxiv.org/abs/2511.04831.

[19] OpenAI. API Pricing and Prompt Caching, 2026. URL https://developers.openai.com/api/docs/pricing.

[20] Anthropic. Claude Opus 5.5: Pricing, 2026. URL https://platform.claude.com/docs/en/models/opus-5-5/overview.

[21] Google. Gemini API Pricing, 2026. URL https://ai.google.dev/gemini-api/docs/pricing.